

### **Final Reflection — THE ARZENS Security Internship**

My experience as a Security Intern at THE ARZENS has been an important part of my technical and professional growth. As a BS Artificial Intelligence student, I joined the program with an interest in technology and AI, but I had limited practical experience in cybersecurity and security automation. Over the course of the internship, I was able to work on practical projects and understand how programming, automation, APIs, AI, and security can be combined to solve real-world problems.

My biggest win was completing practical security projects and turning what I learned into working tools. One of the projects I found particularly valuable was CyberGuard AI, where I worked with Python and Streamlit to build a security analysis application. I also gained experience with threat intelligence, API integration, log analysis, automation workflows, AI-assisted security tools, cloud automation, and dashboards. Working on these projects helped me move beyond simply learning concepts and actually apply them in projects that could be demonstrated and documented.

My biggest challenge was working with technologies and workflows that were initially unfamiliar to me. At the beginning, concepts such as APIs, automation workflows, cloud services, and security tooling were new and sometimes difficult to understand. Instead of avoiding these challenges, I learned to break problems into smaller tasks, research what I did not understand, test my work, and fix errors step by step. This gradually made me more comfortable with troubleshooting and working independently.

Technically, I improved my Python, API integration, security automation, threat intelligence, documentation, testing, and project organization skills. I also learned that building a project involves much more than writing code. Documentation, testing, usability, organization, and being able to explain how a system works are equally important.

Professionally, the internship improved my ability to manage tasks, communicate technical work, document progress, and complete projects within requirements. It also gave me a clearer understanding of the type of work involved in cybersecurity and security engineering.

My advice to future interns is to not be discouraged when a technology seems difficult at first. Start with the basics, work on one problem at a time, ask questions when necessary, and keep testing your work. Practical experience becomes much more valuable when you take the time to understand what you are building rather than only trying to complete the task. This internship showed me that consistent learning and hands-on practice can turn unfamiliar concepts into useful skills.